Assignment 1

1. Let H be a finite non-empty subset of a group G. If H is closed under multiplication then H is a subgroup of G.

2. Show that a subgroup of a cyclic group is cyclic.

3. Let G be a finite cyclic goup of order n. Show that for every divisor d of n, G has a unique subgroup of order d.

4. Compute:
   a) $Hom(\mathbb{Z}|_{4\mathbb{Z}}, \mathbb{Z}|_{6\mathbb{Z}})$.

   b) $Hom(\mathbb{Z}|_{4\mathbb{Z}}, \mathbb{Z}|_{7\mathbb{Z}})$

   c) $Hom(\mathbb{Z}|_{4\mathbb{Z}}, \mathbb{Z})$

5. Show that a group of order 4 is abelian.